

Statistical computing

unit 1: Mathematical Logic Introduction:

- conjunction ($\wedge$)
 - Disjunction ($\vee$)
 - negation ($\sim$)
- } truth table, formula & statement of problems.
De Morgan's law

- conditional Statement ($\rightarrow$)
- Biconditional Statement ($\leftrightarrow$)

- Tautology
 - contradiction
- $(P \sim P = P \vee \sim P)$
 $(P \sim P = P \wedge \sim P)$

• Laws: • Sets

• Subsets

- operation on sets: • union, • intersection
 - complement, • Difference
- Set laws: ~~also~~ associative law. ($\cup, \cap, \sim, \Delta$)

• ~~Dia graph~~ • Venn diagram.

unit 3 & unit 4

• Arithmetic mean

• Direct method $\Rightarrow \bar{x} = \frac{\sum fx}{\sum f}$

• Derived method $\Rightarrow$

$$\bar{x} = a + \frac{\sum fd}{\sum f}$$

• Step deviation method $\Rightarrow$

$$\bar{x} = a + \frac{c(\sum fd)}{\sum f}$$

- standard deviation (σ) $\Rightarrow \sigma = C \sqrt{\left\{ \frac{\sum fd^2}{Ef} - \left(\frac{\sum fd}{Ef} \right)^2 \right\}}$
- variance $\Rightarrow (SD)^2$

- coefficient of variation $\Rightarrow$ Arithmetic mean $\Rightarrow \bar{x} = a + c \left(\frac{\sum fd}{Ef} \right)$
Standard deviation = σ

$$CV = \frac{\text{Standard deviation}}{\text{Arithmetic mean}} \times 100 \Rightarrow \frac{\sigma}{\bar{x}} \times 100$$

- calculate mode: $\rightarrow z = l + \left\{ \frac{(f - f_1)C}{2f - f_1 - f_2} \right\}$

- calculate median: $M = \left[\frac{N+1}{2} \right]^{\text{th}}$

$$M = l + \left\{ \frac{\left(\frac{N}{2} - m \right) C}{f} \right\}$$

Defination: • coefficient of variance

• set, subset

• Simple linear regression

• cor-relation (& its types)

- Define & explain types of Statement (conjunctive, disjunctive)

• Set operations

• Define mean, median, mode

• Define Standard Deviation

• what is statistics & Explain (every basic thing)

• rank of cor-relation: $\sim$ example as question $R = 1 - 6 \left(\frac{\sum d^2}{n^3 - n} \right)$

• co-rank of cor-relation

• Frequency distribution

unit: 2

• Properties of Relation:

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- Reflexive
- Symmetric
- Transitive

• Matrices Addition & multiplication, exponent, inverse.

• Directed graphs.

• Cartesian product,

• Determination of relation eg: $\exists BUC$, $\forall x(B \cap C)$ etc.

Demorgan's law: i) $(A \cup B)' = A' \cap B'$

ii) $(A \cap B)' = A' \cup B'$

Associative law: $(A \cap B) \cap C = A \cap (B \cap C)$